

# Final Project Report

## Concept

Our game is a 3D Bullet Hell RPG with roguelike elements. Here we have a player which will progress through 3 levels. When the player starts off and when they beat a level, they get rewarded a skill/item point, which awards them with 3 choices, either upgrading a stat, or unlocking an ability. While we do use the general trope of a bullet hell, the genre itself is a space mostly dominated by 2d games, with few 3d, and we add on top of that a couple elements such as a parkour level.

## Visual and Artistic Elements

All elements we used are either standard unity elements, or from approved Kenney kits. To add on top of the kits, we added in visual and auditory feedback by adding a visual damage effect and noise when enemies get hit, which involves blending the base texture with the predetermined color of red. The player's health is represented through a static health bar on a unity canvas, and taking damage causes the bar to shake and smoothly decrease. Additionally we added in a fade in and out effect for the walls that block the camera view of the player. The artistic style is also reflected in the sounds as we tried to find sounds that seem to blend with the more simple style of our art compared to a realistic type of look. Colors throughout the game share a similar highly saturated look that works to reflect the same appearance of the assets. This makes a cohesive visual environment where it all seems to belong together.

## Algorithmic Complexity

We have a couple parts of the game that utilize what we learned in class, such as Object Pooling, orbiting saws, items, etc. The Object pooling groups bullets that our player shoots to save on memory instead of consistently destroying and initializing the game object, as that is a costly operation especially when scaled up. Enemy bullet hits are simply calculated using colliders, using triggers to determine when a tagged collider (bullet) intersects with the player collider. We considered alternatives, like stretching bullet colliders to cover the distance traveled between frames, but since enemy bullets needed to be slow enough to dodge, standard colliders worked the best. Items use scriptable objects to create a system that supports adding stats to the player and abilities. The final boss's second phase involves spawning minions in front of it and utilizes some mathematics to do so. `Random.insideUnitCircle` is used to generate a point in a circle of radius one and uses those points to offset the spawn location for the mobs that are spawned in by the boss. The fade effect that can be seen when a wall is blocking the view of the player involves raycasting a ray to determine whether there is obstruction between the ray and the player then calling scripts to lower the alpha of the material and anything that is connected to the wall.

# Gameplay

In terms of player controls, we chose to follow the intuitive control scheme shared with other games, namely fps games. WASD is used to move around, left shift to dash, and left mouse click to shoot. In addition, the player is able to control their orientation alongside the camera by moving their mouse or trackpad around. We initially wanted to add camera smoothing for a more visually polished feel, but decided the snappiness of immediate aiming was more important for gameplay, especially in a shooter game. Throughout the game, the player is given the opportunity to increase their base stats or gain new abilities by picking up gems. Gems allow for variety between runs as well as making the player feel rewarded for completing the levels. The gameplay for our demo introduces 3 levels. The first level is a stage with enemies that shoot and advance towards the player. The player may not be familiar with the game mechanics or controls, so enemies are predictable and less punishing. The second level is a bonus parkour level, which the player can complete to obtain a gem or opt to skip out on the extra power. The level itself is designed to require the player to use their dash to get across platforms, in case they haven't had to during the first stage. Providing the player with this optional challenge also allows them to manage their ideal difficulty for the rest of the run. These first two levels give the player an opportunity to become more comfortable with shooting and movement, preparing them for the upcoming boss. The third and final level is where the player is tasked with defeating the boss and ultimately winning the game. The boss is designed to be much more challenging, using complex bullet patterns and having much more health than the beginning enemies. This introduces a new higher level of challenge to the player, to keep pace with their newfound comfortability with controls and collected powerups.

# Completeness

The game provides a main menu with options to adjust volume of sounds and music of the game, and a controls button that gives the player the button layout for controls. The player can then play through the game with multiple levels and rewards, as described in the gameplay section. The first two levels build upon each other and effectively work as tutorials until the player reaches the third stage boss. Defeating the boss will result in a victory screen, but dying before then will result in the death screen and getting sent back to the main menu.